

Electrical Circuits I

ENGR 2405

Purna Murthy

Chapter 04 – Circuit Theorems

Austin Community College

Overview

- Superposition
- Source transformation
- Thevenin and Norton's theorems
- Applications

Superposition

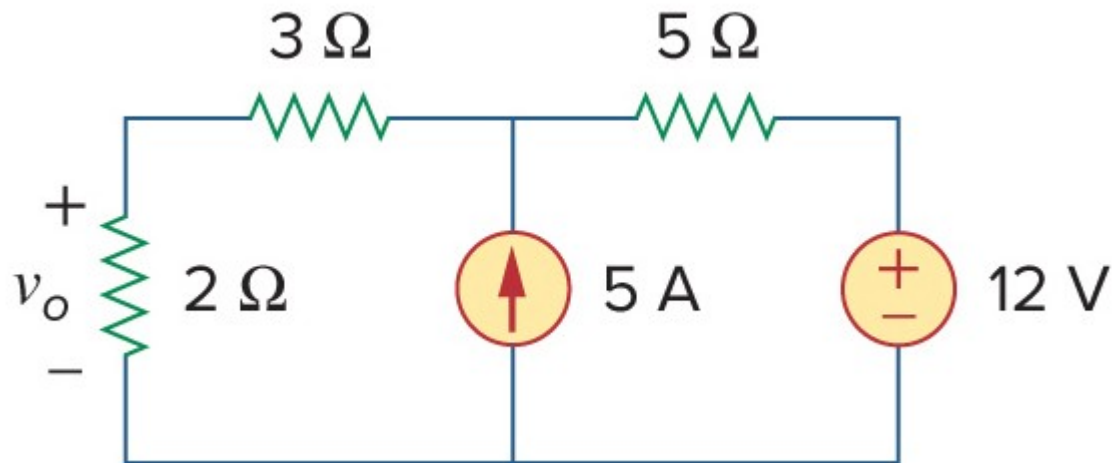
- Superposition is another approach to circuits
- Only works on linear circuits
- The superposition principle states that the voltage across an element in a circuit is the sum of the voltages across that element due to each independent source acting alone
- Also applies to currents through an element

Applying Superposition

- Using superposition means applying one independent source at a time
- Dependent sources are left alone
- The steps are:
 1. Turn off all independent sources except one source
 2. Find the output (V or I) due to that active source
 3. Repeat 1 and 2 for each independent source
 4. Sum the results

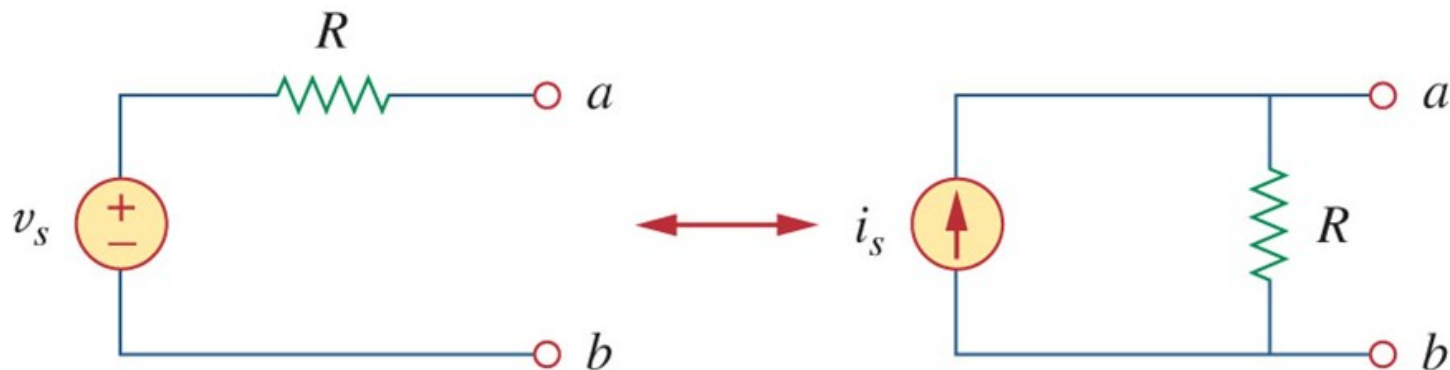
Practice Problems

4.3 Using superposition, find v_o



Source Transformation

- You can transform a source from one form to another (similar to Δ -Y transformation) - this can be useful for simplifying circuits
- A source transformation is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa



Equivalency and Rules

- This works because the two sources have equivalent behavior at their terminals
- With sources off, $R_{ab} = R$
- Short a-b should have same I, thus:

$$v_s = i_s R \text{ or } i_s = \frac{v_s}{R}$$

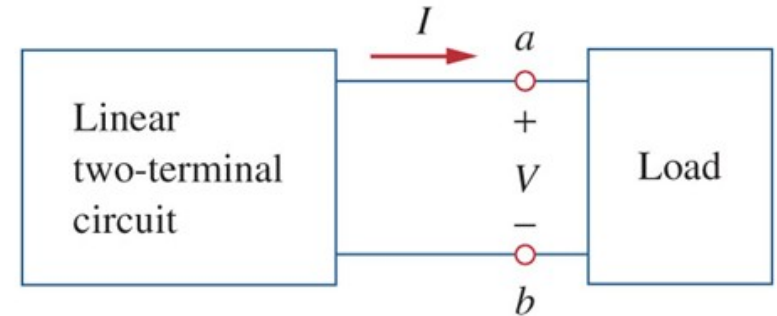
- Dependent sources can also be transformed, but the dependent variable must be handled carefully
- Arrow of the I source points toward V+
- If $R=0$ or $R=\infty$, transformation is not possible (ideal sources)

Thevenin's Theorem

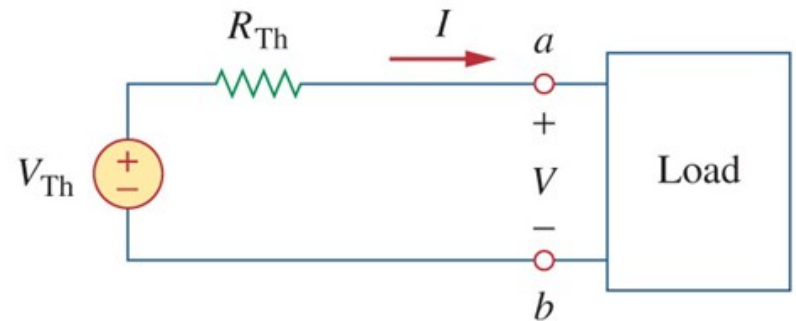
- In many circuits, only one element will be variable
- An example of this is mains power - many different appliances may be plugged into the outlet, each presenting a different resistance
- This variable element is called the load
- Ordinarily one would have to reanalyze the circuit for each change in the load

Thevenin's Theorem

- Thevenin's theorem states that a linear two terminal circuit may be replaced with a voltage source and resistor
- The voltage source's value is equal to the open circuit voltage at the terminals
- The resistance is equal to the resistance measured at the terminals when the independent sources are turned off



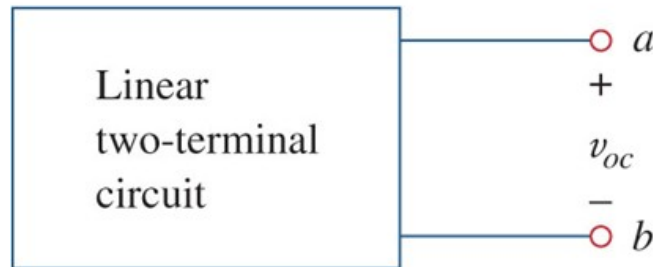
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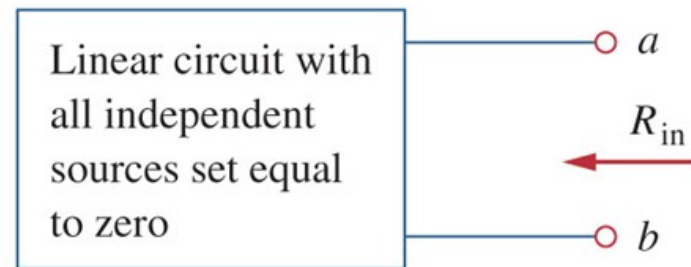
Thevenin's Theorem

- There are two cases to consider when finding the equivalent resistance
- Case 1: If there are no dependent sources, then the resistance may be found by simply turning off all the sources



$$V_{Th} = v_{oc}$$

(a)

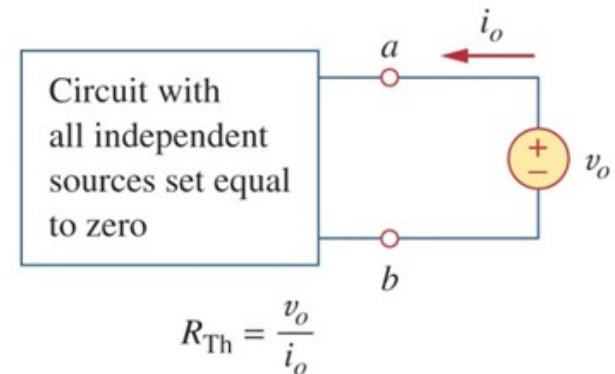


$$R_{Th} = R_{in}$$

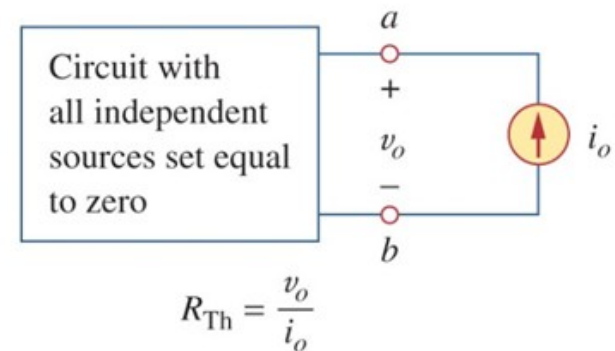
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Thevenin's Theorem

- Case 2: If there are dependent sources, we still turn off all the independent sources
- Now apply a voltage v_o (or current i_o) to the terminals and determine the current i_o (voltage v_o)



(a)



(b)

Thevenin's Theorem

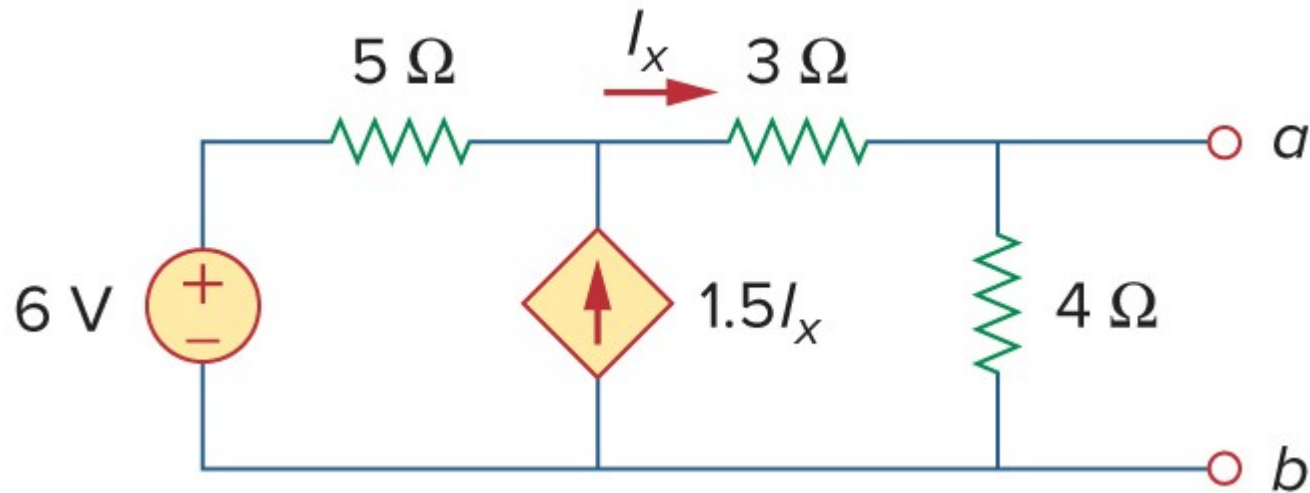
- Thevenin's theorem is very powerful in circuit analysis as it allows one to simplify a circuit
- A large circuit may be replaced by a single independent voltage source and a single resistor
- The equivalent circuit behaves externally exactly the same as the original circuit

Negative Resistance?

- It is possible for the result of this analysis to end up with a negative resistance
- This implies the circuit is supplying power
- This is reasonable with dependent sources
- Note that in the end, the Thevenin equivalent makes working with variable loads much easier

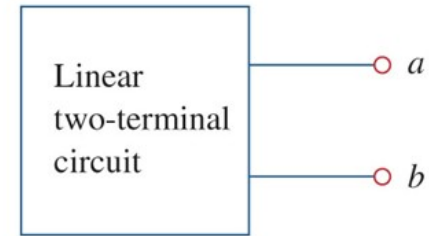
Practice Problems

4.9 Find the Thevenin equivalent

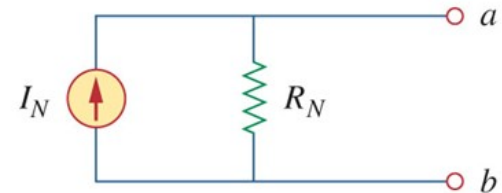


Norton's Theorem

- Similar to Thevenin's theorem, Norton's theorem states that a linear two terminal circuit may be replaced with an equivalent circuit containing a resistor and a current source
- The Norton resistance will be exactly the same as the Thevenin



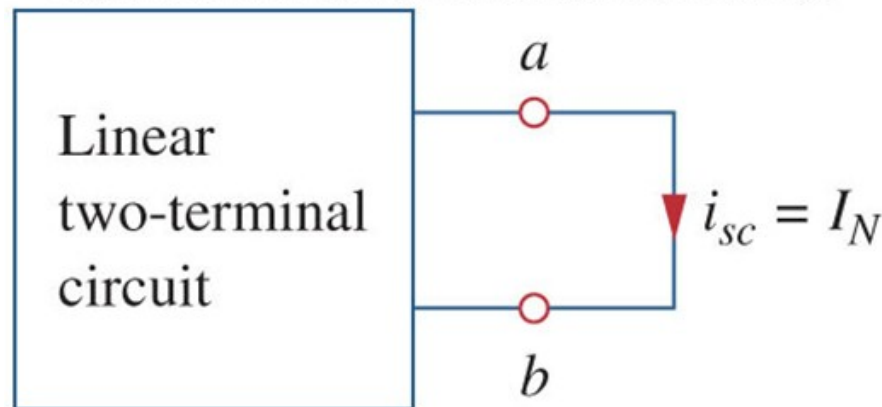
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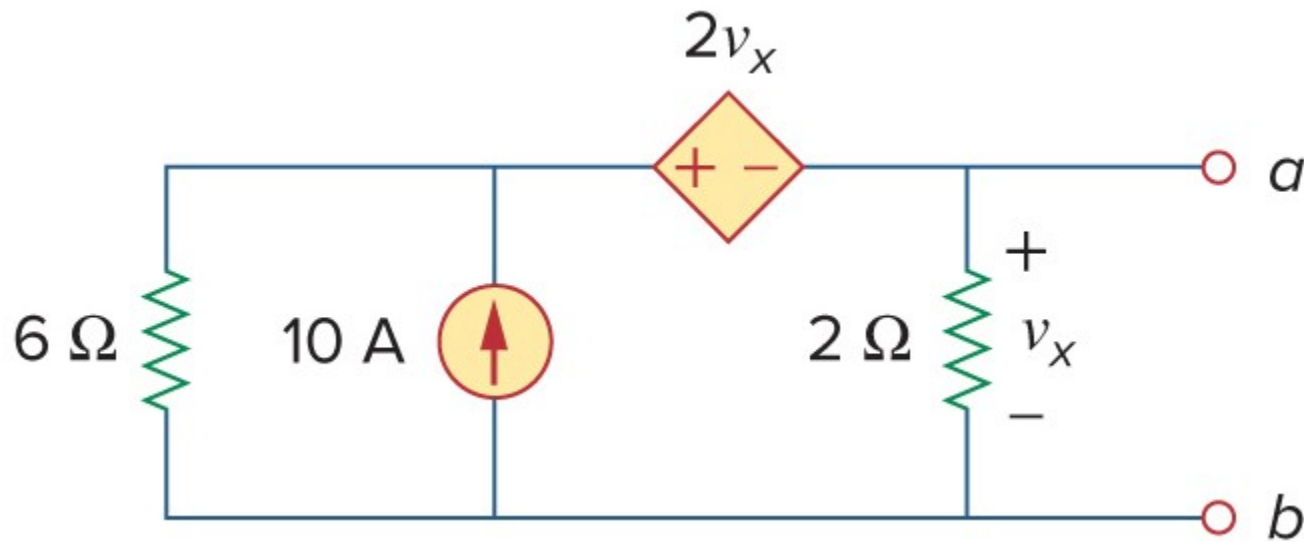
Norton's Theorem

The Norton current I_N is found by short circuiting the circuit's terminals and measuring the resulting current, $I_N = i_{sc}$



Practice Problems

4.12 Find the Norton equivalent



Norton versus Thevenin

- These two equivalent circuits can be related to each other via source transformation:

$$I_N = \frac{V_{Th}}{R}$$

- To find these equivalents, need 2 of the following:
 - V_{ab} (open-circuit voltage)
 - I_{sc} (short-circuit current)
 - R_{ab} (input resistance at a-b when independent sources are turned off)

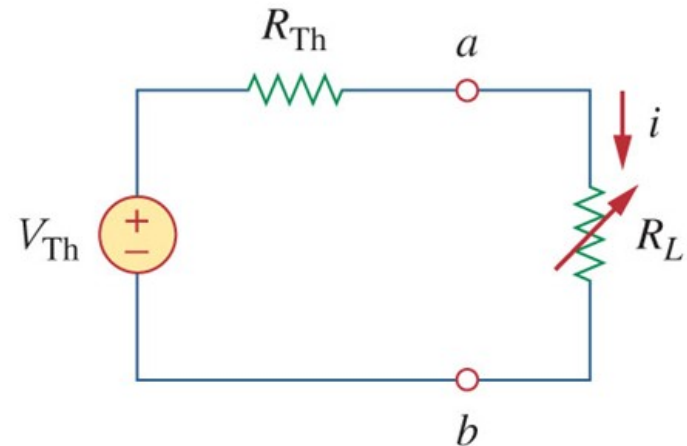
Maximum Power Transfer

- In many applications, a circuit is designed to power a load
- Among those applications there are many cases where we wish to maximize the power transferred to the load
- Unlike an ideal source, internal resistance will restrict the conditions where maximum power is transferred

Maximum Power Transfer

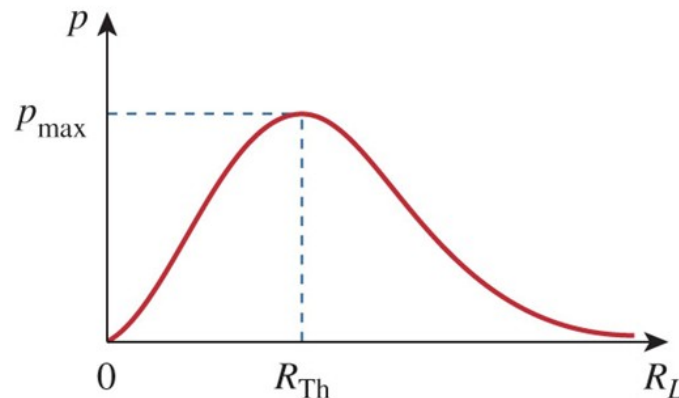
- We can use the Thevenin equivalent circuit for finding the maximum power in a linear circuit
- We will assume that the load resistance can be varied
- Looking at the equivalent circuit with load included, the power transferred is:

$$p = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$



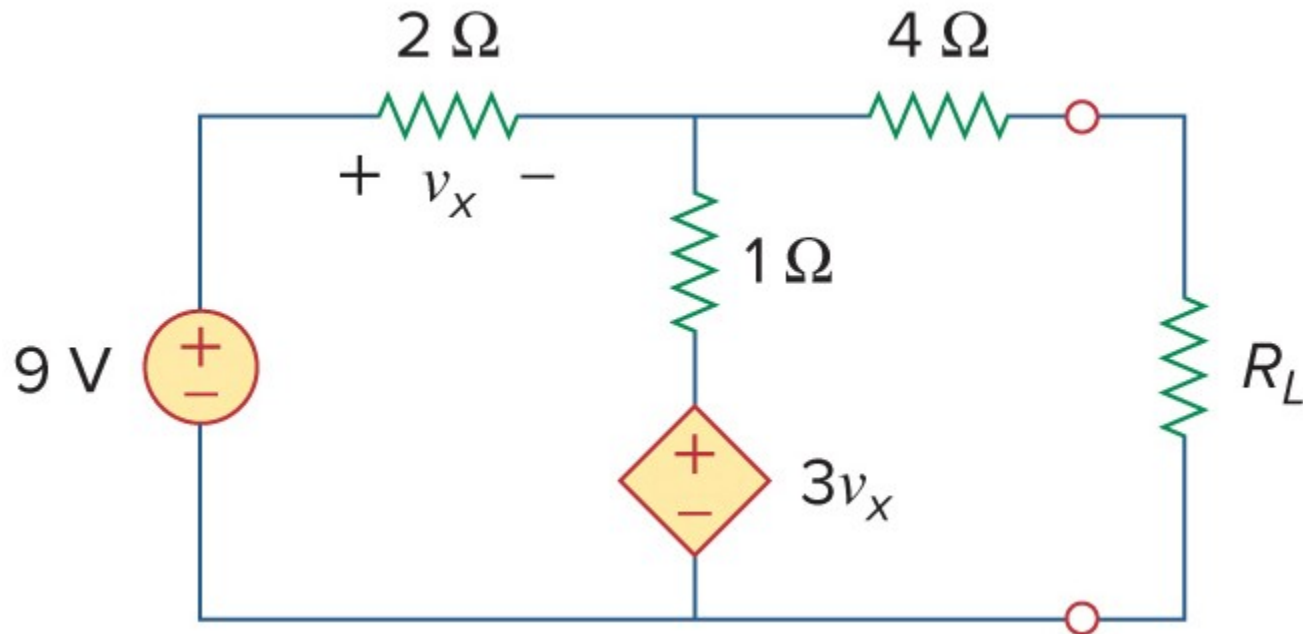
Maximum Power Transfer

- For a given circuit, V_{TH} and R_{TH} are fixed
- By varying the load resistance R_L , the power delivered to the load varies as shown
- You can see that as R_L approaches 0 and ∞ the power transferred goes to zero
- In fact the maximum power transferred is when $R_L = R_{TH}$



Practice Problems

4.13 Determine R_L for max power transfer

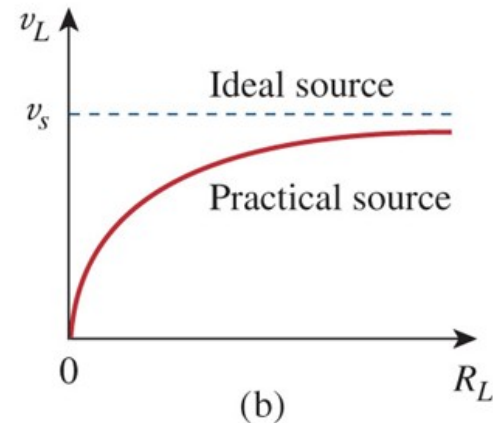
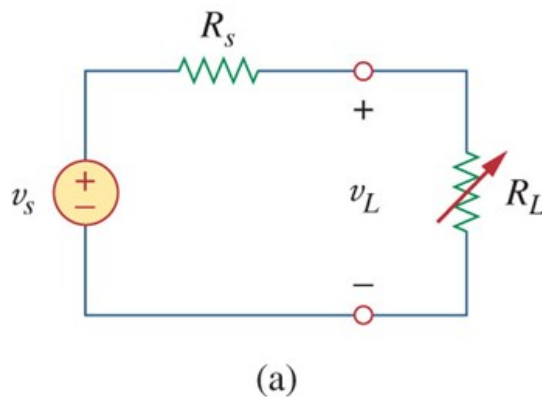


Realistic Sources

- The Thevenin and Norton equivalent circuits are useful in understanding the behavior of realistic sources
- Recall that ideal sources have $0/\infty$ internal resistance
- The Thevenin and Norton circuits introduce deviations from these ideals

Source Modeling

- Take the Thevenin circuit with load resistor:
 - The internal resistor and the load act a voltage divider
 - The lower the load resistance, the more voltage drop that occurs in the source

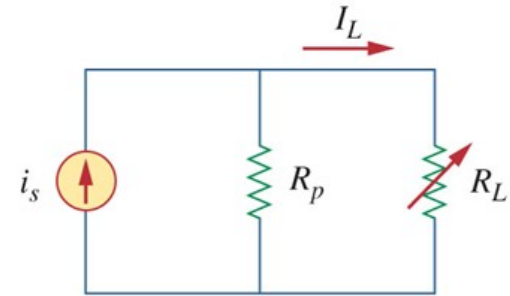


Source Modeling

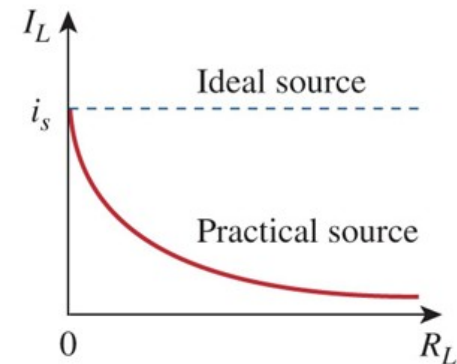
- This means that as the load resistance increases, the voltage source comes closer to operating like the ideal source
- Similarly, with a realistic current source, the internal resistor in parallel with the ideal source acts to siphon away current that would otherwise go to the load

Source Modeling

- Here, the load and the internal resistor act as a current divider
- From that perspective, the lower the load resistance, the more current passes through it
- Thus lower load resistance leads to behavior closer to the ideal source



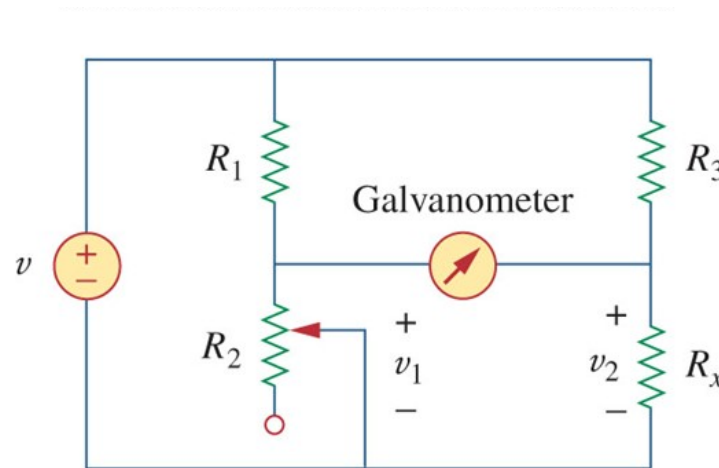
(a)



(b)

Resistance Measurement

- Although the ohmmeter is the most common method for measuring resistance, there is a more accurate method
- It is called the Wheatstone bridge
- It is based on the principle of the voltage divider



Resistance Measurement

- Using three known resistors and a galvanometer, an unknown resistor can be tested.
- The unknown resistor is placed at the position R_4
- The variable resistor R_2 is adjusted until the galvanometer shows zero current
- At this point, the bridge is “balanced” and the voltages from the two dividers are equal

Balanced Bridge

- When balanced, the unknown resistor's value is:

$$R_x = \frac{R_3}{R_1} R_2$$

- The key to the high accuracy lies in the fact that any slight difference in the voltage dividers will lead to a current flow

$$I = \frac{V_{TH}}{R_{TH} + R_x}$$

- Where the bridge, less the unknown resistor, is reduced to its Thevenin equivalent